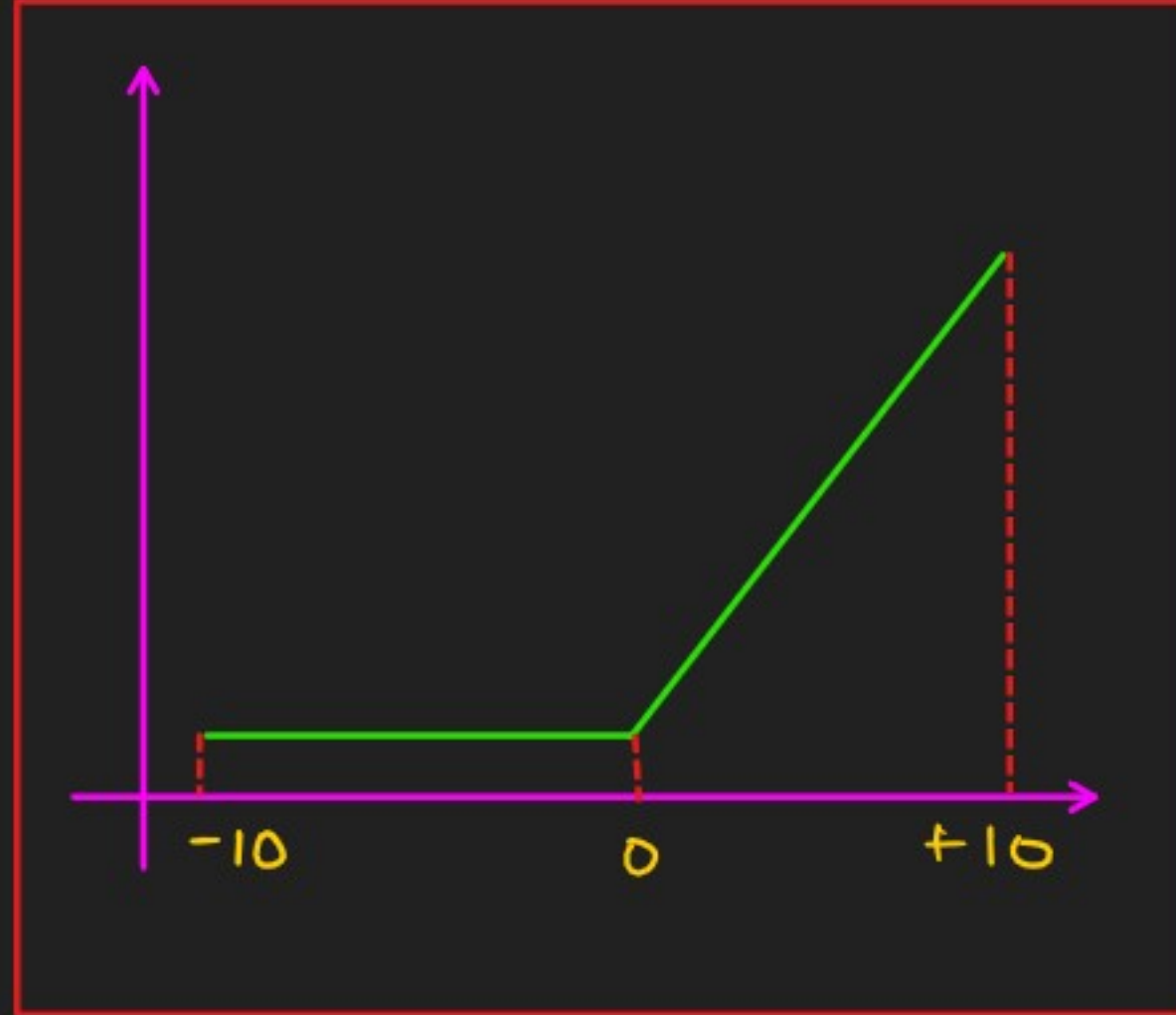


Lecture-21: GELU Activation Function

- We will implement a small neural network sub-module that is a part of the LLM transformer block
- Two activation function commonly implemented in LLMs.
 - GELU $\rightarrow x \cdot \phi(x)$ $\rightarrow$ CDF of standard Gaussian distribution
 - SwiGLU

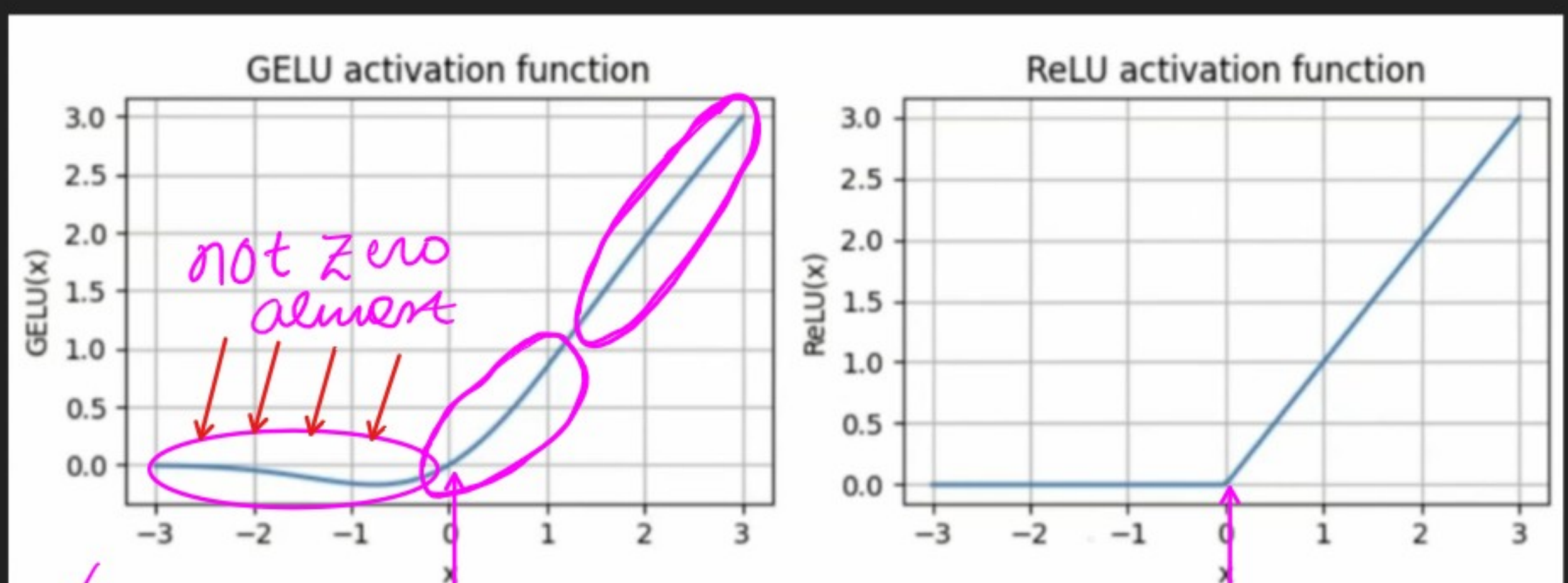


ReLU Activation Function

- Here is the GELU activation function approximate used for training GPT-2.

$$\text{GELU}(x) \approx 0.5 \cdot x \cdot (1 + \tanh[\sqrt{2/\pi} \cdot (x + 0.44715 \cdot x^3)])$$

- let's code now



- prevent dead neuron problem

- at $x=0$ differentiable
- It's seems to work better ReLU experiment with LLM.

- at $x=0$ non-differentiable

① Using the GELU activation function, we implement the small neural network module. Feed Forward that we will be using the LLM transformer block.

② let us code

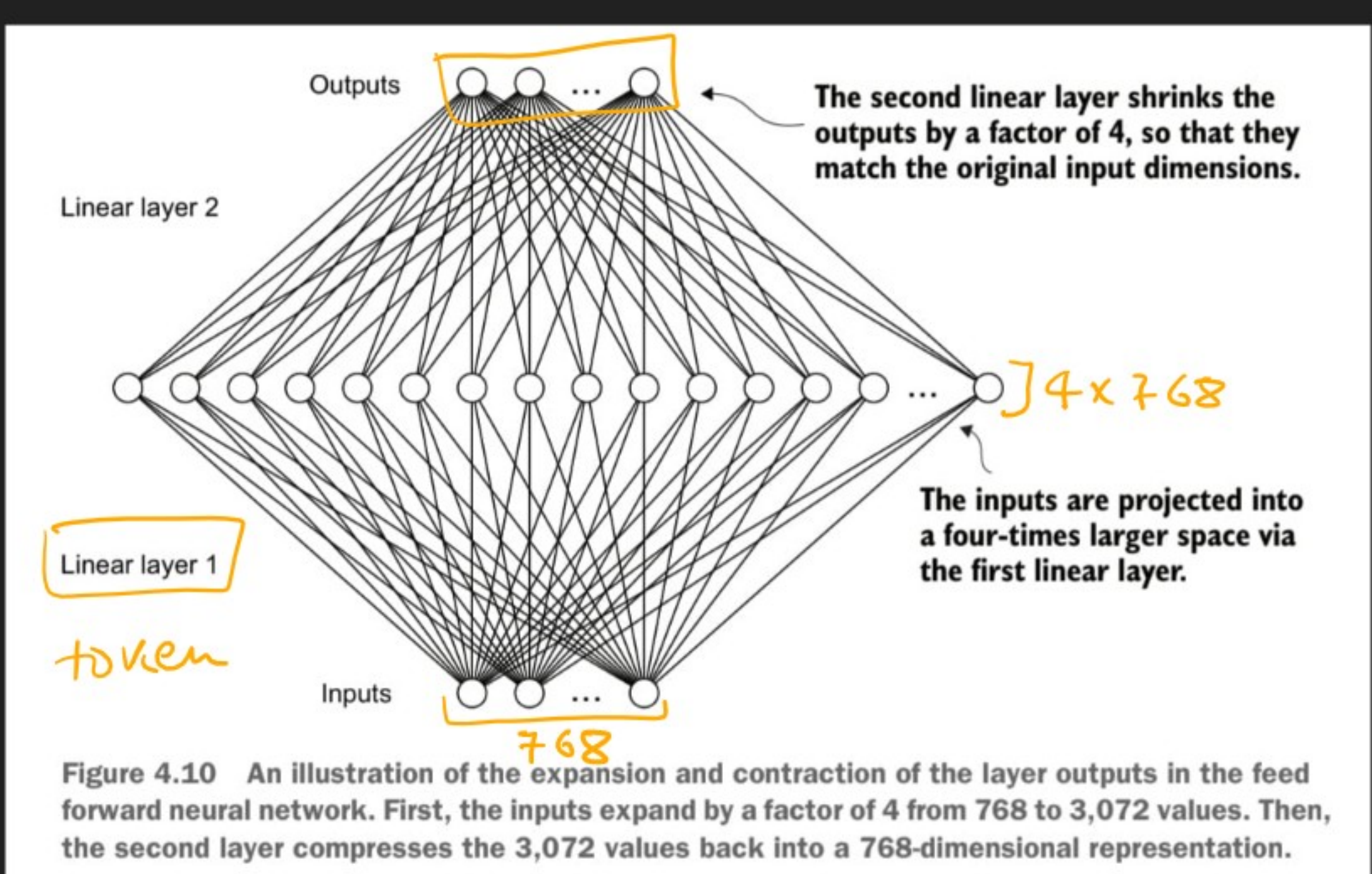


Figure 4.10 An illustration of the expansion and contraction of the layer outputs in the feed forward neural network. First, the inputs expand by a factor of 4 from 768 to 3,072 values. Then, the second layer compresses the 3,072 values back into a 768-dimensional representation.

expansion and contraction allows for a rich exploration space.

